

Template-Free Brain MRI Alignment Using Outlier-Robust Wasserstein Distance under Rigid Transformation

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Abstract—Brain MRI alignment is a fundamental topic in neuroscience. However, brain images collected by medical devices are usually noisy, and it is hard to guarantee that all the subjects maintain consistent physical postures during the collection processes; moreover, existing alignment methods often rely on pre-specified templates. To address these challenges, first, we propose a novel alignment method based on *outlier-Robust Wasserstein Distance under Rigid transformation (RWD)*. Our method enjoys several advantages, such as it can capture the geometric structure for alignment under rigid transformation, and the outlier robustness can effectively mitigate the impact of noise. Then, we extend our method to compute the corresponding outlier-robust barycenter for group-level analysis. Notably, our approaches for computing pairwise alignments and barycenters are both template-free. Finally, we conduct a set of experiments on brain MRI to evaluate the effectiveness of our alignment method and barycenter algorithm. The results suggest that our approach outperforms the state-of-the-art methods in handling noisy images under rigid transformation.

Index Terms—Outlier-robust Wasserstein distance, Rigid transformation, Brain alignment, Barycenter.

I. INTRODUCTION

Recently, humans have benefited a lot from millimeter anatomical and functional brain imaging. Nevertheless, these images exhibit high variability in anatomo-functional structures [1]. Even for an individual patient, his brain has different structures across different pathological stages [2], [3]. Such variability hinders us from deriving generalizable conclusions from medial images.

To address the variability problem, several alignment methods have been proposed. The popular Multimodal Surface Matching (MSM) [4], [5] aligns functional information after introducing functional maps into the diffeomorphic framework; the heuristics *hyperalignment* [6]–[8] methods align brains by using cortical patches. Additionally, some methods [1], [9], [10] align brains based on Wasserstein distance by modeling brain images as probability measures.

Except for the variability problem, brain images collected by medical devices, such as Functional Magnetic Resonance Imaging (fMRI), are often noisy. As a popular non-invasive imaging technique for recording brain activity, fMRI images are susceptible to various “artifacts” (or “noise”), such as physiological noise [11], motion artifacts [12], and magnetic field inhomogeneity [13]. Furthermore, during image collection processes, it is challenging to guarantee that all subjects maintain consistent physical postures; even the same subject may adopt different postures for different image collection processes. Moreover, existing alignment methods often rely on pre-specified templates; it can be quite challenging if the template is unavailable when studying new species. In conclusion, existing methods [1], [4]–[9] lack robustness to noise and rigid transformations, and often depend on pre-specified templates.

Additionally, individual brains often exhibit substantial deviations in characteristics. To derive generalizable conclusions for a species or group [14]–[17], we often need to compute the barycenter (i.e., average brain) at the group level.

Contributions: To tackle these challenges, **firstly**, we propose an alignment method based on *outlier-Robust Wasserstein Distance [18] under Rigid transformation (RWD)*. Our method inherits the ability of Wasserstein distance for capturing geometry structures and thus can capture anatomo-functional structures well, which makes it resilient to local deformations; the outlier robustness mitigates the impact of noise; its invariance under rigid transformation accommodates variations in physical postures across subjects. **Then**, based on our alignment method, we compute the corresponding outlier-robust barycenter for group-level analysis. Notably, our approaches for computing pairwise alignments and barycenters

do not rely on any pre-specified templates¹. **Finally**, experimental results show that our method can obtain high-quality alignments on noisy datasets; its invariance under rigid transformation frees subjects from maintaining consistent physical postures during the image collection processes. Moreover, the corresponding barycenter algorithm provides finer details for group-level analysis.

II. PRELIMINARIES

A. Notations

Let $[n] := \{1, \dots, n\}$ and $\|\cdot\|_p$ be the ℓ_p -norm of a vector. Let $(\mathcal{X}, \text{dist})$ be a metric space. Let $\mathbb{R}_+$ be the set of non-negative real numbers. We use $\mathcal{P}(\mathcal{X})$ to denote the probability measure space on $\mathcal{X}$. All rigid transformations in d -dimensional Euclidean spaces $\mathbb{R}^d$ form a mathematical group called Euclidean group $E(d)$. We denote matrices by capital boldface letters, such as $\mathbf{C}$; C_{ij} is its element in the i -th row and j -th column. Similarly, vectors are represented by lowercase boldface letters, such as $\mathbf{a} := (a_1, \dots, a_d)^T \in \mathbb{R}^d$; a_i is its i -th element; $\mathbf{a} \succeq \mathbf{b}$ means $a_i \geq b_i$ for all $i \in [d]$. We define $(\mathbf{a}; \mathbf{b}) := (a_1, \dots, a_d, b_1, \dots, b_d)^T$ for any $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$.

Additionally, we define two discrete probability measures

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i}, \beta = \sum_{j=1}^n b_j \delta_{y_j} \in \mathcal{P}(\mathcal{X}), \quad (1)$$

where $\mathbf{a}, \mathbf{b} \in \mathbb{R}_+^n$ are their weights and $\{x_i\}_{i \in [n]}, \{y_j\}_{j \in [n]} \subseteq \mathcal{X}$ their locations; δ is the Dirac delta function.

The Wasserstein distance [19] is a popular tool for measuring the similarity between two probability measures. However, its sensitivity to noise restricts its applicability in many real-world scenarios; thus, we introduce its robust version to alleviate the impact of outliers.

Definition II.1 (Outlier-robust Wasserstein distance [18]). Given α, β as in (1), $0 \leq \zeta_\alpha, \zeta_\beta < 1, z \geq 1$, and a cost matrix $\mathbf{C} \in \mathbb{R}_+^{n \times n}$ with $C_{ij} = \text{dist}^z(x_i, y_j)$, the outlier-robust Wasserstein distance between α and β is

$$W(\alpha, \beta) := \min_{\mathbf{P} \in \Pi(\alpha, \beta)} \langle \mathbf{P}, \mathbf{C} \rangle,$$

where $\Pi(\alpha, \beta) = \{\mathbf{P} \in \mathbb{R}_+^{n \times n} \mid \|\mathbf{a}'\|_1 = \zeta_\alpha, \|\mathbf{b}'\|_1 = \zeta_\beta, \mathbf{P}\mathbf{1} = \frac{\mathbf{a}-\mathbf{a}'}{1-\zeta_\alpha}, \mathbf{P}^T\mathbf{1} = \frac{\mathbf{b}-\mathbf{b}'}{1-\zeta_\beta}, \mathbf{a}', \mathbf{b}' \in \mathbb{R}_+^n\}$, and $\mathbf{1}$ is the vector of ones.

Remark II.2. $\mathbf{a}', \mathbf{b}'$ are the weights of outliers for α, β , and $\zeta_\alpha, \zeta_\beta$ are their total mass respectively. The $\Pi(\alpha, \beta)$ is the set of all feasible coupling. In the following, we design our method in a popular metric space, the Euclidean space $\mathbb{R}^d$.

III. OUR METHODS

In this section, we first model brain images as probability measures, and then define the RWD for pairwise alignments. Next, we give the algorithm for solving RWD. Further, we compute the corresponding barycenter.

¹Our method can operate directly on voxels, thus it is template-free. For the template-dependent case, we can fix rigid transformation as identity transformation, which is much simpler than the template-free case.

A. Modeling brains as probability measures

An individual source brain is usually represented by a set of vertexes $\{x_i\}_{i \in [n]} \subseteq \mathbb{R}^d$. Each vertex x_i is associated with both locations (i.e., anatomical information) $\bar{x}_i \in \mathbb{R}^{\bar{d}}$ and features (i.e., functional information) $\hat{x}_i \in \mathbb{R}^{\hat{d}}$ with $d = \bar{d} + \hat{d}$; more specifically, given a pre-specified θ with $0 \leq \theta \leq 1$, we define $x_i = (\theta \cdot \bar{x}_i; (1-\theta) \cdot \hat{x}_i) \in \mathbb{R}^d$, and model the source brain as a measure α as follows

$$\alpha = \sum_{i=1}^n a_i \cdot \delta_{(\theta \cdot \bar{x}_i; (1-\theta) \cdot \hat{x}_i)} = \sum_{i=1}^n a_i \cdot \delta_{x_i} \in \mathcal{P}(\mathbb{R}^d). \quad (2)$$

Similarly, we define $y_j = (\theta \cdot \bar{y}_j; (1-\theta) \cdot \hat{y}_j) \in \mathbb{R}^d$, and model the target brain as a measure β as follows

$$\beta = \sum_{j=1}^n b_j \cdot \delta_{(\theta \cdot \bar{y}_j; (1-\theta) \cdot \hat{y}_j)} = \sum_{j=1}^n b_j \cdot \delta_{y_j} \in \mathcal{P}(\mathbb{R}^d). \quad (3)$$

B. Pairwise alignments

For a pre-specified $0 \leq \theta \leq 1$, we define RWD as²

$$\mathcal{W}(\alpha, \beta) := \min_{\mathbf{P} \in \Pi(\alpha, \beta), e \in E(\bar{d})} \theta \cdot \underbrace{\langle \mathbf{P}, e \circ \bar{\mathbf{C}} \rangle}_{\text{term-1}} + (1-\theta) \cdot \underbrace{\langle \mathbf{P}, \hat{\mathbf{C}} \rangle}_{\text{term-2}} \quad (4)$$

to measure the similarity between α and β , where e is rigid transformation, $\bar{C}_{ij} = \|\bar{x}_i - \bar{y}_j\|_2^2$, $\hat{C}_{ij} = \|\hat{x}_i - \hat{y}_j\|_2^2$ and $(e \circ \bar{C})_{ij} = \|e \circ \bar{x}_i - \bar{y}_j\|_2^2$. The coupling $\mathbf{P}$ induced by $\mathcal{W}(\alpha, \beta)$ gives the alignment relationship between the source brain and the target brain. The **term-1** maintains the anatomical structures, the **term-2** preserves the feature information, and θ makes a trade-off between them.

C. Algorithm for RWD

The aim of Algorithm 1 is to find a measure $\tilde{\alpha} \in \mathcal{P}(\mathcal{X})$ such that $\mathcal{W}(\alpha, \beta) = W(\tilde{\alpha}, \beta)$. We solve $\mathcal{W}(\alpha, \beta)$ by updating the coupling $\mathbf{P}$ and rigid transformation e alternatively. First, we fix e , and have

$$\begin{aligned} \mathcal{W}_e(\alpha, \beta) &:= \min_{\mathbf{P} \in \Pi(\alpha, \beta)} \langle \mathbf{P}, \theta \cdot (e \circ \bar{\mathbf{C}}) + (1-\theta) \cdot \hat{\mathbf{C}} \rangle \\ &= \min_{\mathbf{P} \in \Pi(\alpha, \beta)} \langle \mathbf{P}, \mathbf{C} \rangle = W(e \circ \alpha, \beta), \end{aligned} \quad (5)$$

where $C_{ij} := \theta \cdot \|e \circ \bar{x}_i - \bar{y}_j\|_2^2 + (1-\theta) \cdot \|\hat{x}_i - \hat{y}_j\|_2^2$ and $e \circ \alpha := \sum_{i=1}^n a_i \cdot \delta_{(\theta \cdot (e \circ \bar{x}_i); (1-\theta) \cdot \hat{x}_i)}$; thus, we can update $\mathbf{P}$ by computing $W(e \circ \alpha, \beta)$. This step can be solved within $\tilde{\mathcal{O}}(\frac{n^2}{\epsilon_+})$ time [20], where ϵ_+ is the pre-specified additive error.

Then, we fix $\mathbf{P}$, and have $\arg \min_e \mathcal{W}_{\mathbf{P}}(\alpha, \beta) = \arg \min_e \langle \mathbf{P}, e \circ \bar{\mathbf{C}} \rangle$. We partition the rigid transformation e into translation transformation e_1 and orthogonal transformation e_2 ; that is, $e = e_2 \circ e_1$. The translation transformation can be updated as $e_1 = \sum_{i=1}^n \sum_{j=1}^n P_{ij} y_j - \sum_{i=1}^n \sum_{j=1}^n P_{ij} x_i$. For fixed $e_1, \mathbf{P}$, computing the optimal orthogonal transformation e_2 is an orthogonal Procrustes problem [21]. More specifically,

²The subscript $*$ in $\mathcal{W}_*(\cdot, \cdot)$ means that we fix the parameter $*$. For example, $\mathbf{P}$ in $\mathcal{W}_{\mathbf{P}}(\cdot, \cdot)$ means that we fix the parameter $\mathbf{P}$.

we first obtain $\mathbf{M} = \sum_{ij} P_{ij} x_i y_j^T$, and apply singular value decomposition $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^T$; then, we have

$$e_2 = \mathbf{U}\mathbf{V}^T. \quad (6)$$

Remark III.1. In our method, the locations of β are fixed, and the rigid transformation is imposed on source α . Moreover, ζ_α means that $\|\mathbf{P}\mathbf{1} - \mathbf{a}\|_1 \leq 2 \cdot \zeta_\alpha$. For the case $\zeta_a = 0, \zeta_b = 0$, $\mathcal{W}(\cdot, \cdot)$ is a (semi-)metric.

Algorithm 1 Algorithm for alignments

Input: $\alpha, \beta, \zeta_\alpha, \zeta_\beta$

- 1: $t = 0$;
- 2: **for** $t < T_{align}$ **do**
- 3: $t = t + 1$;
- 4: **▷ update coupling P**
Obtain coupling $\mathbf{P}$ by computing $W(e \circ \alpha, \beta)$ according to (5);
- 5: **▷ update translation transformation e_1 and orthogonal transformation e_2**
Compute $e_1 = \sum_{i=1}^n \sum_{j=1}^n P_{ij} y_j - \sum_{i=1}^n \sum_{j=1}^n P_{ij} x_i$;
Compute $e_2 = \mathbf{U}\mathbf{V}^T$ according to (6), and set $e = e_2 \circ e_1$;
- 6: **end for**

Output: $\mathbf{P}, \tilde{\alpha}$

D. Algorithm for barycenter

Let

$$\begin{aligned} \mathcal{Q} &= \left\{ \alpha^k = \sum_{i=1}^n a_i^k \cdot \delta_{(\theta \cdot \bar{x}_i^k; (1-\theta) \cdot \hat{x}_i^k)} \right. \\ &= \left. \sum_{i=1}^n a_i^k \cdot \delta_{x_i^k} \in \mathcal{P}(\mathbb{R}^d) \right\}_{k \in [m]} \end{aligned} \quad (7)$$

be a set of m measures with ζ_α being their mass of outliers. The barycenter on $\mathcal{Q}$ is a measure β (as in (3)) with ζ_β being its mass of outliers, and satisfies

$$\beta = \arg \min_{\beta \in \mathcal{P}(\mathbb{R}^d)} \sum_{k=1}^m \mathcal{W}(\alpha^k, \beta). \quad (8)$$

To simplify the algorithm for computing barycenter, we pre-specified the weights and locations of the barycenter. Thus, we only need to optimize its features. In the following, we compute the barycenter β by updating the couplings $\mathbf{P}^k, k \in [m]$ and the features $\{\hat{y}_j\}_{j \in [n]}$ alternatively. More specifically, we first compute the $\mathbf{P}^k, \tilde{\alpha}^k$ with $(\alpha^k, \beta, \zeta_\alpha, \zeta_\beta)$ as input by Algorithm 1, and set $\alpha^k = \tilde{\alpha}^k$. Then, the features of β are computed by

$$\hat{y}_j = \sum_{k=1}^m \sum_{i=1}^n P_{ij}^k \hat{x}_{ij}^k / \left(\sum_{k=1}^m \sum_{i=1}^n P_{ij}^k \right). \quad (9)$$

The detailed procedures are in Algorithm 2.

The β_0 can be obtained by selecting the $\alpha \in \mathcal{Q}$ that minimize $\sum_{k=1}^m \mathcal{W}(\alpha^k, \alpha)$. Actually, we can also initialize β_0

Algorithm 2 Algorithm for barycenter

Input: $\mathcal{Q} = \{\alpha^k\}_{k \in [m]}, \beta_0, \zeta_\alpha, \zeta_\beta$

- 1: $t = 0, \beta = \beta_0$;
- 2: **for** $t < T_{BC}$ **do**
- 3: $t = t + 1$;
- 4: **▷ update coupling $\mathbf{P}^k$.**
For each $\alpha^k, k \in [m]$, compute $\mathbf{P}^k, \tilde{\alpha}^k$ by Algorithm 1 with $(\alpha^k, \beta, \zeta_\alpha, \zeta_\beta)$ as input;
Set $\alpha^k = \tilde{\alpha}^k$;
- 5: **▷ update features**
For each $\hat{y}_j, j \in [n]$, compute $\hat{y}_j$ by using (9);
 $\beta = \sum_{j=1}^n b_j \cdot \delta_{(\theta \cdot \bar{y}_j; (1-\theta) \cdot \hat{y}_j)}$;
- 6: **end for**

Output: β

by selecting a measure from $\mathcal{Q}$ randomly. If the number n of voxels for the images is large, we can improve the scalability by clustering algorithms. Specifically, we gather the voxels into several clusters, represent each cluster by its center, and assign the size of the cluster as its weight; that is, we replace images with a set of weighted centers.

IV. EXPERIMENTS

This section demonstrates the effectiveness of our method for computing pairwise alignments and barycenter. All experiments are performed on a server equipped with 2.40GHz Intel CPU, NVIDIA GeForce RTX 3090 GPU, 64GB main memory, and Python 3.12. Our RWD is executed on the CPU, while FUGW runs on the GPU. Our implementation utilizes some Python libraries [22]–[24].

Dataset: The Individual Brain Charting (IBC)³ [25], [26] is a high spatial-resolution, multi-task fMRI dataset. We select 13 subjects from IBC, and 433 fMRI maps for each subject; We call the above data an original dataset $\mathcal{Q}$. To generate a noisy dataset $\mathcal{Q}'$, we add $\zeta = \zeta_\alpha = \zeta_\beta$ mass of artifacts and apply a random rotation to each original image. The proper margin violation is at most 2ζ .

A. Pairwise alignments

Evaluation metrics: (i) $\text{diff}_a := \|\mathbf{P}\mathbf{1} - \mathbf{a}\|_1$ records the deviation of coupling $\mathbf{P}$ from margin $\mathbf{a}$. (ii) $\text{diff}_b := \|\mathbf{P}^T \mathbf{1} - \mathbf{b}\|_1$ records the deviation of coupling $\mathbf{P}$ from margin $\mathbf{b}$. (iii) loss_{align} refers to the average distance between pairwise noisy data.

To demonstrate the performance of our pairwise alignment method, we applied our algorithms to the noisy dataset $\mathcal{Q}'$. Each case involves 13 subjects. Thus, we compute $\frac{13 \times 12}{2} = 78$ pairwise alignments between the subjects. The recorded values are the average results for these 78 pairwise alignments. The hyperparameter ρ of FUGW⁴ controls the robustness to noise.

³Its original [25] and extended version [26] includes 12 and 15 subjects, respectively.

⁴Code and additional details for FUGW are available on GitHub: ic-casp2025.

Table I shows that our RWD has better alignment quality than FUGW on noisy datasets, and it is robust to rigid transformations. Moreover, our method does not cause excessive margin violation (i.e., $\text{diff}_a = \text{diff}_b \leq 2\zeta$).

TABLE I: Comparisons of different alignment methods on the noisy datasets. WD is a special case of RWD, where rigid transformation is replaced with identity transformation.

ζ	method	ρ	time(s)	diff_a	diff_b	$\text{loss}_{\text{align}}(\downarrow)$
0.1	FUGW	0.50	1627	0.28	0.27	153.14 ± 63.16
	FUGW	1	1626	0.21	0.21	154.91 ± 63.65
	FUGW	2	1626	0.16	0.16	157.64 ± 62.84
	WD	None	328	0.20	0.20	3.35 ± 1.12
	RWD	None	1475	0.20	0.20	2.42 ± 0.50
0.2	FUGW	0.13	2020	0.44	0.45	157.83 ± 64.94
	FUGW	0.25	2021	0.33	0.34	154.73 ± 67.28
	FUGW	0.50	2023	0.26	0.27	157.49 ± 66.87
	WD	None	398	0.40	0.40	2.72 ± 1.05
	RWD	None	1746	0.40	0.40	2.01 ± 0.54
0.3	FUGW	0.03	2651	0.62	0.64	204.80 ± 94.36
	FUGW	0.06	2651	0.58	0.59	196.32 ± 93.91
	FUGW	0.13	2652	0.46	0.48	186.11 ± 92.22
	WD	None	487	0.59	0.59	3.84 ± 2.39
	RWD	None	2152	0.59	0.59	1.74 ± 0.73

TABLE II: Comparisons of different barycenter algorithms on noisy datasets.

ζ	method	ρ	time(s)	diff	$\text{loss}(\downarrow)$
0.1	FUGW	20	569	0.24	67.22 ± 26.85
	FUGW	24	569	0.20	67.07 ± 27.07
	FUGW	28	569	0.17	67.46 ± 26.84
	WD	None	125	0.20	68.37 ± 28.54
	RWD	None	467	0.20	8.43 ± 0.17
0.2	FUGW	12	640	0.48	23.80 ± 6.01
	FUGW	16	640	0.35	23.58 ± 5.72
	FUGW	20	642	0.27	23.45 ± 5.84
	WD	None	139	0.40	19.20 ± 5.52
	RWD	None	502	0.40	1.10 ± 0.11
0.3	FUGW	8	732	0.82	132.55 ± 39.10
	FUGW	12	732	0.56	117.68 ± 36.00
	FUGW	16	732	0.40	111.21 ± 34.96
	WD	None	148	0.60	94.73 ± 32.63
	RWD	None	562	0.60	1.19 ± 0.12

B. Barycenter

Evaluation metrics: (i) diff records the average deviation of couplings from the weights of m input data and the barycenter. (ii) loss represents the cost incurred from the barycenter to the clean data.

Table II shows the results of our barycenter algorithm. We select six data from the noisy dataset $\mathcal{Q}'$ as input. We compare our barycenter method (RWD-BC) with its corresponding version without rigid transformation (WD-BC) and the FUGW-based method (FUGW-BC). Table II shows that our RWD-BC yields a higher quality barycenter across varying levels of noise.

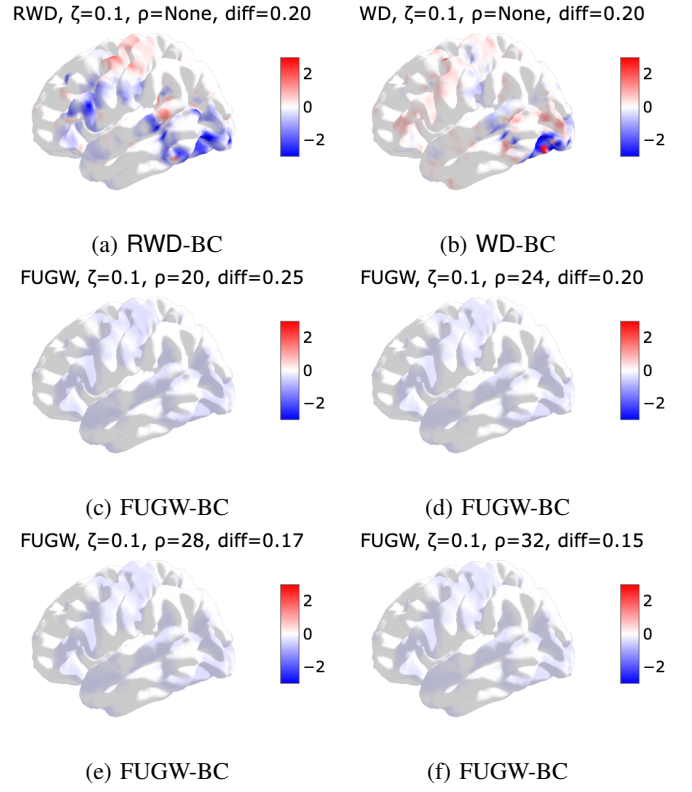


Fig. 1: z-score of barycenters.

Figure 1 illustrates the z-scores of the barycenters. Compared to FUGW-BC and WD-BC, our RWD-BC captures finer details. The FUGW-BC, which incorporates an entropy regularization term, exhibits a diffusion-like behavior when applied to voxel fMRI images (as seen in Figure 1). In FUGW [1], this diffusion effect is mitigated by mapping voxel images onto a pre-specified template before calculating the barycenter, making the method dependent on pre-specified template. In contrast, our method does not include the entropy regularization term, allowing it to directly handle voxel images without the need for pre-specified templates.

In conclusion, our method achieves better alignment quality and produces a finer barycenter without relying on any pre-specified template. Moreover, it eliminates the need for tuning the parameter ρ to handle varying levels of noise and does not result in excessive marginal violations. Additionally, our algorithm is more computationally efficient on a CPU than FUGW on a GPU, saving both time and hardware resources.

V. CONCLUSION

This paper proposes an RWD-based alignment method and its corresponding barycenter algorithm. It can tackle outliers efficiently and accommodate variations in physical postures across subjects during image collection. The template-free nature makes it a promising approach, especially for studying new species.

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VI. FUGW

Fused Unbalanced Gromov Wasserstein (FUGW) is an approach for aligning pairwise brains based on optimal transformation. We rewrite the Equation (1) of FUGW in [1] as

$$\begin{aligned} \text{FUGW}(\alpha, \beta) := & \theta \cdot L_W(\mathbf{P}) + (1 - \theta) \cdot L_{GW}(\mathbf{P}) + \rho_1 \cdot KL(\mathbf{P}\mathbf{1}|\boldsymbol{\omega}^s) \\ & + \rho_2 \cdot KL(\mathbf{P}^T\mathbf{1}|\boldsymbol{\omega}^t) + \epsilon \cdot E(\mathbf{P}), \end{aligned} \quad (10)$$

where the term $L_W(\mathbf{P})$ maintains the functional information, the $L_{GW}(\mathbf{P})$ preserves the anatomical structure, and θ makes a trade-off between anatomical structure and functional information; moreover, ρ_1, ρ_2 is used to penalize the margin violations, and $E(\mathbf{P})$ is the entropy regularization term. In our experiments, we set $\rho = \rho_1 = \rho_2$.

Its pairwise alignment between α and β is the coupling $\mathbf{P}$ induced by $\text{FUGW}(\alpha, \beta)$. And its corresponding barycenter on $\{\alpha^k\}_{k \in [m]}$ is a measure β that minimizes the following objective

$$\sum_{k=1}^m \text{FUGW}(\alpha^k, \beta). \quad (11)$$

VII. FULL EXPERIMENTS

This section demonstrates the effectiveness of our method for computing pairwise alignments and barycenter. All experiments are performed on a server equipped with 2.40GHz Intel CPU, NVIDIA GeForce RTX 3090 GPU, 64GB main memory, and Python 3.12. Our RWD is executed on the CPU, while FUGW runs on the GPU. Our implementation utilizes some Python libraries [22]–[24].

IBC Dataset: The Individual Brain Charting (IBC)⁵ [25], [26] is a high spatial-resolution, multi-task fMRI dataset. We select 13 subjects from IBC, and 433 fMRI maps for each subject;

ADHD Dataset: Attention Deficit Hyperactivity Disorder (ADHD) dataset is a widely used multi-site neuroimaging dataset. Due to its multi-site nature, the dataset exhibits substantial variability. Thus, it is commonly used to evaluate the robustness of brain analysis. s

We call the above data an original dataset $\mathcal{Q}$. To generate a noisy dataset $\mathcal{Q}'$, we add $\zeta = \zeta_\alpha = \zeta_\beta$ mass of artifacts and apply a random rotation to each original image. The proper margin violation is at most 2ζ .

A. Pairwise alignments

Evaluation metrics: (i) $\text{diff}_a := \|\mathbf{P}\mathbf{1} - \mathbf{a}\|_1$ records the deviation of coupling $\mathbf{P}$ from margin $\mathbf{a}$. (ii) $\text{diff}_b := \|\mathbf{P}^T\mathbf{1} - \mathbf{b}\|_1$ records the deviation of coupling $\mathbf{P}$ from margin $\mathbf{b}$. (iii) loss_{align} refers to the average distance between pairwise noisy data.

To demonstrate the performance of our pairwise alignment method, we applied our algorithms to the noisy dataset $\mathcal{Q}'$. Each case involves 13 subjects. Thus, we compute $\frac{13 \times 12}{2} = 78$ pairwise alignments between the subjects. The recorded values are the average results for these 78 pairwise alignments. The hyperparameter ρ of FUGW⁶ controls the robustness to noise.

Table III shows that our RWD has better alignment quality than FUGW on noisy datasets, and it is robust to rigid transformations. Moreover, our method does not cause excessive margin violation (i.e., $\text{diff}_a = \text{diff}_b \leq 2\zeta$).

⁵Its original [25] and extended version [26] includes 12 and 15 subjects, respectively.

⁶Code and additional details for FUGW are available on GitHub: iccasp2025.

TABLE III: Comparisons of different alignment methods on noisy dataset. WD is a special case of RWD, where rigid transformation is replaced with identity transformation.

ζ	method	ρ	time(s)	diff _a	diff _b	loss _{align} (↓)
0.1	FUGW	0.02	1626	0.40	0.41	156.70 $\pm$ 61.85
	FUGW	0.03	1626	0.57	0.55	150.55 $\pm$ 59.62
	FUGW	0.06	1626	0.59	0.58	147.28 $\pm$ 59.99
	FUGW	0.13	1616	0.50	0.47	150.64 $\pm$ 59.71
	FUGW	0.25	1623	0.37	0.37	151.63 $\pm$ 61.80
	FUGW	0.50	1627	0.28	0.27	153.14 $\pm$ 63.16
	FUGW	1	1626	0.21	0.21	154.91 $\pm$ 63.65
	FUGW	2	1626	0.16	0.16	157.64 $\pm$ 62.84
	WD	None	328	0.20	0.20	3.35 $\pm$ 1.12
0.2	RWD	None	1475	0.20	0.20	2.42 $\pm$ 0.50
	FUGW	0.02	2029	0.52	0.47	172.96 $\pm$ 67.21
	FUGW	0.03	2023	0.60	0.60	167.30 $\pm$ 67.81
	FUGW	0.06	2021	0.56	0.56	162.62 $\pm$ 68.59
	FUGW	0.13	2020	0.44	0.45	157.83 $\pm$ 64.94
	FUGW	0.25	2021	0.33	0.34	154.73 $\pm$ 67.28
	FUGW	0.50	2023	0.26	0.27	157.49 $\pm$ 66.87
	FUGW	1	2021	0.20	0.21	160.16 $\pm$ 66.20
	FUGW	2	2020	0.14	0.16	159.28 $\pm$ 66.72
0.3	WD	None	398	0.40	0.40	2.72 $\pm$ 1.05
	RWD	None	1746	0.40	0.40	2.01 $\pm$ 0.54
	FUGW	0.02	2650	0.52	0.52	208.50 $\pm$ 92.33
	FUGW	0.03	2651	0.62	0.64	204.80 $\pm$ 94.36
	FUGW	0.06	2651	0.58	0.59	196.32 $\pm$ 93.91
	FUGW	0.13	2652	0.46	0.48	186.11 $\pm$ 92.22
	FUGW	0.25	2652	0.33	0.35	177.40 $\pm$ 91.42
	FUGW	0.50	2653	0.24	0.27	173.17 $\pm$ 91.25
	FUGW	1	2651	0.18	0.21	172.34 $\pm$ 91.35
0.3	FUGW	2	2644	0.13	0.17	173.02 $\pm$ 91.79
	WD	None	487	0.59	0.59	3.84 $\pm$ 2.39
	RWD	None	2152	0.59	0.59	1.74 $\pm$ 0.73

Figure 2 presents the visualization of different alignment methods with $\zeta = 0.1$ on IBC dataset. It can be observed that our RWD-based alignment method achieves better performance. This advantage is particularly evident in the rightmost column.

Specifically, WD-based methods fails to effectively suppress noise, indicating limited robustness to noise. The FUGW-based approach needs search the hyperparameter ρ , which is difficult to tune in practice for varying mass of noise; moreover, the diffusion effect introduced by this method leads to noticeable information degradation.

In contrast, our RWD-based alignment method achieves more accurate alignments while preserving more details, demonstrating superior robustness and stability under noisy conditions.

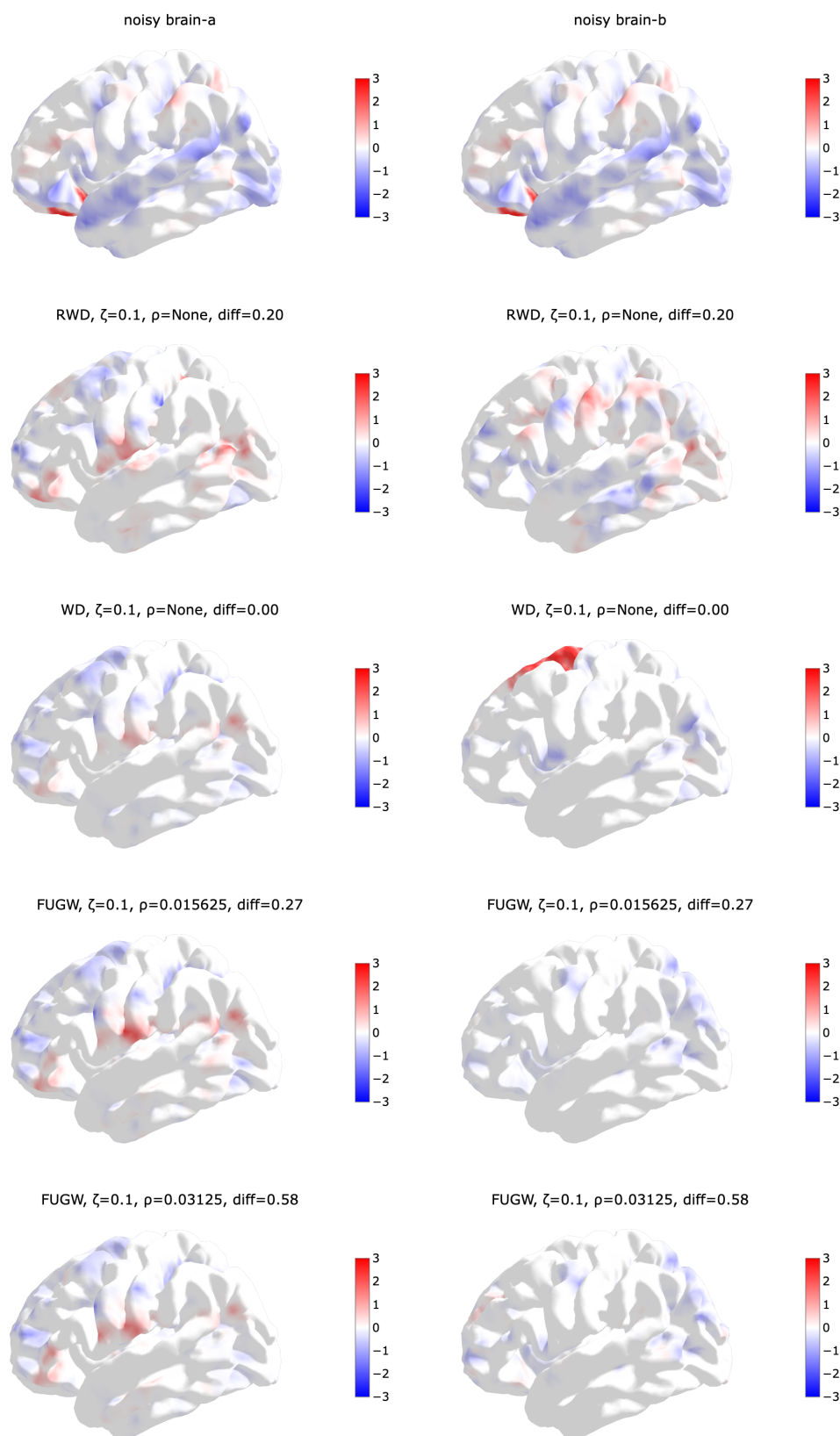


Fig. 2: Visualization of alignment methods with $\zeta = 0.1$ on IBC dataset.

TABLE IV: Comparisons of different barycenter algorithms on noisy IBC dataset.

ζ	method	ρ	time(s)	diff	loss($\downarrow$)
0.1	FUGW	4	571	1.11	66.70 $\pm$ 26.62
	FUGW	8	571	0.64	67.28 $\pm$ 26.66
	FUGW	12	573	0.42	67.09 $\pm$ 26.64
	FUGW	16	572	0.31	67.52 $\pm$ 26.79
	FUGW	20	569	0.24	67.22 $\pm$ 26.85
	FUGW	24	569	0.20	67.07 $\pm$ 27.07
	FUGW	28	569	0.17	67.46 $\pm$ 26.84
	FUGW	32	572	0.15	67.66 $\pm$ 26.84
	FUGW	36	572	0.14	67.43 $\pm$ 27.01
	FUGW	40	572	0.12	67.44 $\pm$ 26.87
	WD	None	125	0.20	68.37 $\pm$ 28.54
	RWD	None	467	0.20	8.43 $\pm$ 0.17
0.2	FUGW	4	640	1.18	24.74 $\pm$ 5.91
	FUGW	8	640	0.73	24.72 $\pm$ 5.98
	FUGW	12	640	0.48	23.80 $\pm$ 6.01
	FUGW	16	640	0.35	23.58 $\pm$ 5.72
	FUGW	20	642	0.27	23.45 $\pm$ 5.84
	FUGW	24	641	0.22	23.61 $\pm$ 5.94
	FUGW	28	641	0.19	23.55 $\pm$ 5.79
	FUGW	32	642	0.16	23.59 $\pm$ 6.03
	FUGW	36	641	0.15	23.71 $\pm$ 5.90
	FUGW	40	640	0.13	23.90 $\pm$ 5.95
	WD	None	139	0.40	19.20 $\pm$ 5.52
	RWD	None	502	0.40	1.10 $\pm$ 0.11
0.3	FUGW	4	731	1.12	160.09 $\pm$ 43.12
	FUGW	8	732	0.82	132.55 $\pm$ 39.10
	FUGW	12	732	0.56	117.68 $\pm$ 36.00
	FUGW	16	732	0.40	111.21 $\pm$ 34.96
	FUGW	20	732	0.31	108.37 $\pm$ 34.58
	FUGW	24	731	0.25	107.41 $\pm$ 34.69
	FUGW	28	732	0.21	106.96 $\pm$ 34.41
	FUGW	32	730	0.18	106.39 $\pm$ 34.50
	FUGW	36	731	0.16	105.28 $\pm$ 34.50
	FUGW	40	732	0.14	105.26 $\pm$ 34.40
	WD	None	148	0.60	94.73 $\pm$ 32.63
	RWD	None	562	0.60	1.19 $\pm$ 0.12

187 0.20 8371 23045 60 0.20 8327 17980

TABLE V: Comparisons of different barycenter algorithms on noisy ADHD dataset.

ζ	method	ρ	time(s)	diff	loss($\downarrow$)
0.2	FUGW	4	1938	0.85	4201590 $\pm$ 229009
	FUGW	12	1724	0.47	6629606 $\pm$ 216254
	FUGW	16	1602	0.32	7785562 $\pm$ 209044
	FUGW	16	1739	0.25	8442489 $\pm$ 205096
	FUGW	20	1559	0.20	8835359 $\pm$ 200088
	FUGW	24	1671	0.17	9131148 $\pm$ 200301
	FUGW	28	1717	0.14	9329023 $\pm$ 197472
	FUGW	32	1736	0.12	9502731 $\pm$ 195819
	FUGW	36	1636	0.11	9619077 $\pm$ 197257
	FUGW	40	1598	0.10	9731046 $\pm$ 193806
	WD	None	74	0.38	7754 $\pm$ 289
0.3	RWD	None	189	0.40	6223 $\pm$ 355
	FUGW	4	1583	0.85	4208795 $\pm$ 228898
	FUGW	8	1386	0.48	6629953 $\pm$ 215310
	FUGW	12	1239	0.32	7787488 $\pm$ 208281
	FUGW	16	1381	0.25	8444297 $\pm$ 205368
	FUGW	20	1555	0.20	8841865 $\pm$ 201912
	FUGW	24	1268	0.17	9129032 $\pm$ 199587
	FUGW	28	1276	0.14	9330253 $\pm$ 198285
	FUGW	32	1243	0.12	9486310 $\pm$ 197715
	FUGW	36	1237	0.11	9615322 $\pm$ 197571
	FUGW	40	1331	0.10	9728489 $\pm$ 195238
0.4	WD	None	69	0.57	12471 $\pm$ 344
	RWD	None	401	0.60	5505 $\pm$ 133
	FUGW	4	1841	0.85	4208651 $\pm$ 29028
	FUGW	8	1580	0.48	6629806 $\pm$ 15218
	FUGW	12	1544	0.32	7787339 $\pm$ 08495
	FUGW	16	1403	0.25	8444145 $\pm$ 05491
	FUGW	20	1417	0.20	8841713 $\pm$ 01806
	FUGW	24	1386	0.17	9128879 $\pm$ 99725
	FUGW	28	1300	0.14	9330099 $\pm$ 98307
	FUGW	32	1349	0.12	9486156 $\pm$ 97716
	FUGW	36	1318	0.11	9615168 $\pm$ 97515
	FUGW	40	1303	0.10	9728335 $\pm$ 95260
	WD	None	62	0.77	49221 $\pm$ 3441
	RWD	None	360	0.79	4648 $\pm$ 235

B. Barycenter

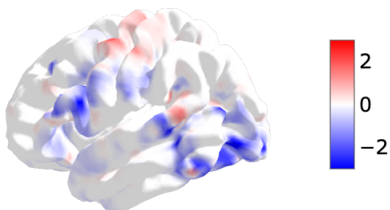
Evaluation metrics: (i) *diff* records the average deviation of couplings from the weights of m input data and the barycenter. (ii) *loss* represents the cost incurred from the barycenter to the clean data.

Table IV shows the results of our barycenter algorithm on IBC dataset. We select six data from the noisy dataset $\mathcal{Q}'$ as input. We compare our barycenter method (RWD-BC) with its corresponding version without rigid transformation (WD-BC) and the FUGW-based method (FUGW-BC). Table II shows that our RWD-BC yields a higher quality barycenter across varying levels of noise.

Figures 3 to 5 illustrate the z-scores of the barycenters on IBC dataset. Compared to FUGW-BC and WD-BC, our RWD-BC captures finer details. The FUGW-BC, which incorporates an entropy regularization term, exhibits a diffusion-like behavior when applied to voxel fMRI images (as seen in Figure 1). In FUGW [1], this diffusion effect is mitigated by mapping voxel images onto a pre-specified template before calculating the barycenter, making the method dependent on pre-specified template. In contrast, our method does not include the entropy regularization term, allowing it to directly handle voxel images without the need for pre-specified templates.

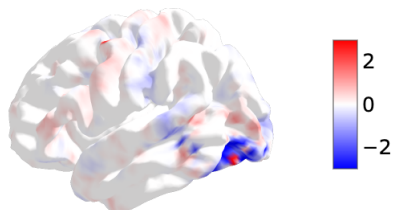
In conclusion, our method achieves better alignment quality and produces a finer barycenter without relying on any pre-specified template. Moreover, it eliminates the need for tuning the parameter ρ to handle varying levels of noise and does not result in excessive marginal violations. Additionally, our algorithm is more computationally efficient on a CPU than FUGW on a GPU, saving both time and hardware resources.

RWD, $\zeta=0.1$, $\rho=\text{None}$, $\text{diff}=0.20$



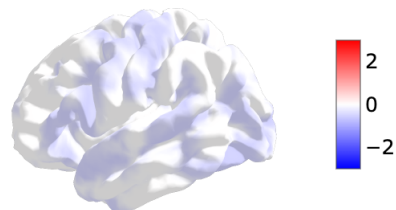
(a) RWD-BC

WD, $\zeta=0.1$, $\rho=\text{None}$, $\text{diff}=0.20$



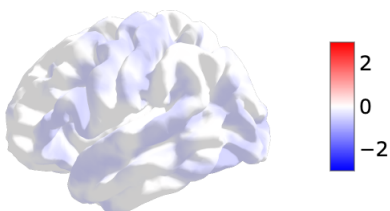
(b) WD-BC

FUGW, $\zeta=0.1$, $\rho=12$, $\text{diff}=0.45$



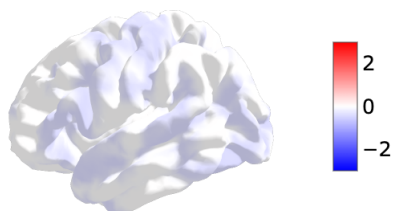
(c) FUGW-BC

FUGW, $\zeta=0.1$, $\rho=16$, $\text{diff}=0.32$



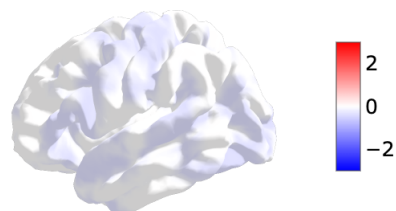
(d) FUGW-BC

FUGW, $\zeta=0.1$, $\rho=20$, $\text{diff}=0.25$



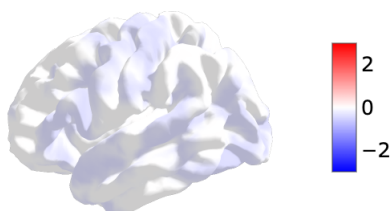
(e) FUGW-BC

FUGW, $\zeta=0.1$, $\rho=24$, $\text{diff}=0.20$



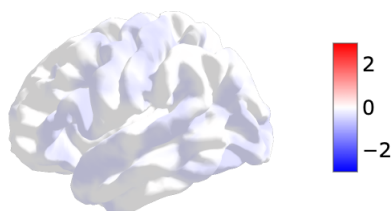
(f) FUGW-BC

FUGW, $\zeta=0.1$, $\rho=28$, $\text{diff}=0.17$



(g) FUGW-BC

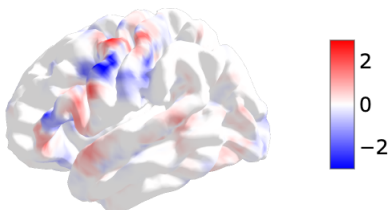
FUGW, $\zeta=0.1$, $\rho=32$, $\text{diff}=0.15$



(h) FUGW-BC

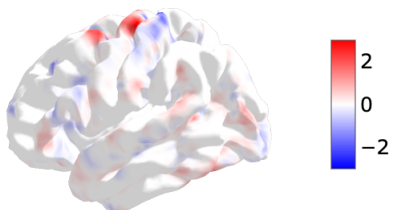
Fig. 3: z-score of barycenters with $\zeta = 0.1$ on IBC dataset.

RWD, $\zeta=0.2$, $\rho=\text{None}$, $\text{diff}=0.40$



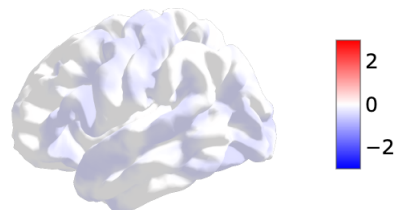
(a) RWD-BC

WD, $\zeta=0.2$, $\rho=\text{None}$, $\text{diff}=0.40$



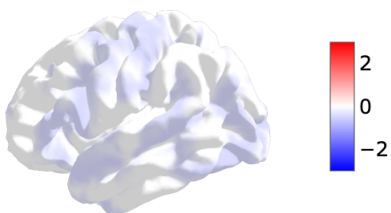
(b) WD-BC

FUGW, $\zeta=0.2$, $\rho=12$, $\text{diff}=0.48$



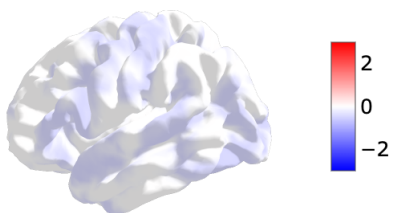
(c) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=16$, $\text{diff}=0.34$



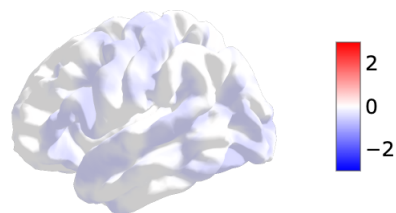
(d) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=20$, $\text{diff}=0.27$



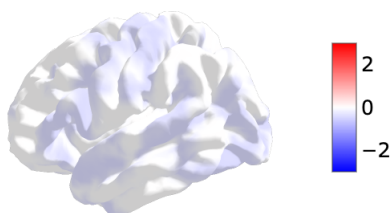
(e) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=24$, $\text{diff}=0.22$



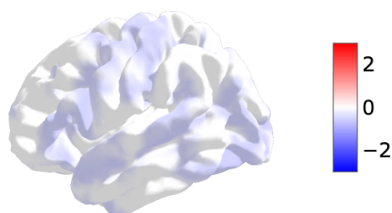
(f) FUGW-BC

FUGW, $\zeta=0.2$, $\rho=28$, $\text{diff}=0.19$



(g) FUGW-BC

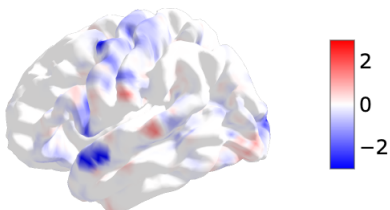
FUGW, $\zeta=0.2$, $\rho=32$, $\text{diff}=0.16$



(h) FUGW-BC

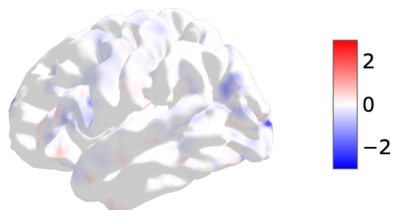
Fig. 4: z-score of barycenters $\zeta = 0.2$ on IBC dataset.

RWD, $\zeta=0.3$, $\rho=\text{None}$, $\text{diff}=0.60$



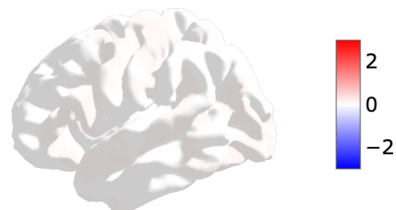
(a) RWD-BC

WD, $\zeta=0.3$, $\rho=\text{None}$, $\text{diff}=0.60$



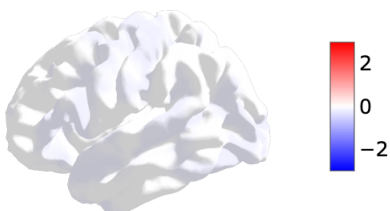
(b) WD-BC

FUGW, $\zeta=0.3$, $\rho=12$, $\text{diff}=0.53$



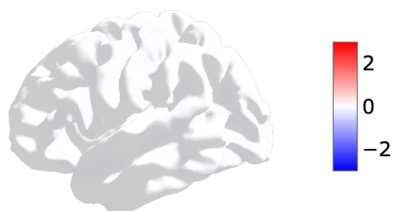
(c) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=16$, $\text{diff}=0.39$



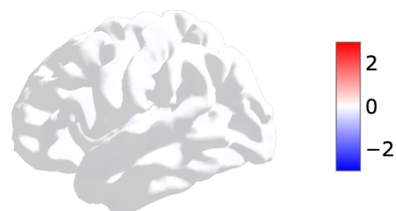
(d) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=20$, $\text{diff}=0.30$



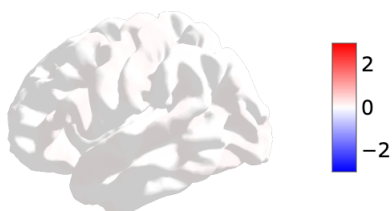
(e) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=24$, $\text{diff}=0.24$



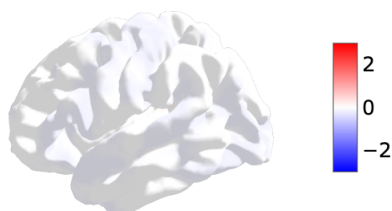
(f) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=28$, $\text{diff}=0.21$



(g) FUGW-BC

FUGW, $\zeta=0.3$, $\rho=32$, $\text{diff}=0.18$



(h) FUGW-BC

Fig. 5: z-score of barycenters $\zeta = 0.3$ on IBC dataset.

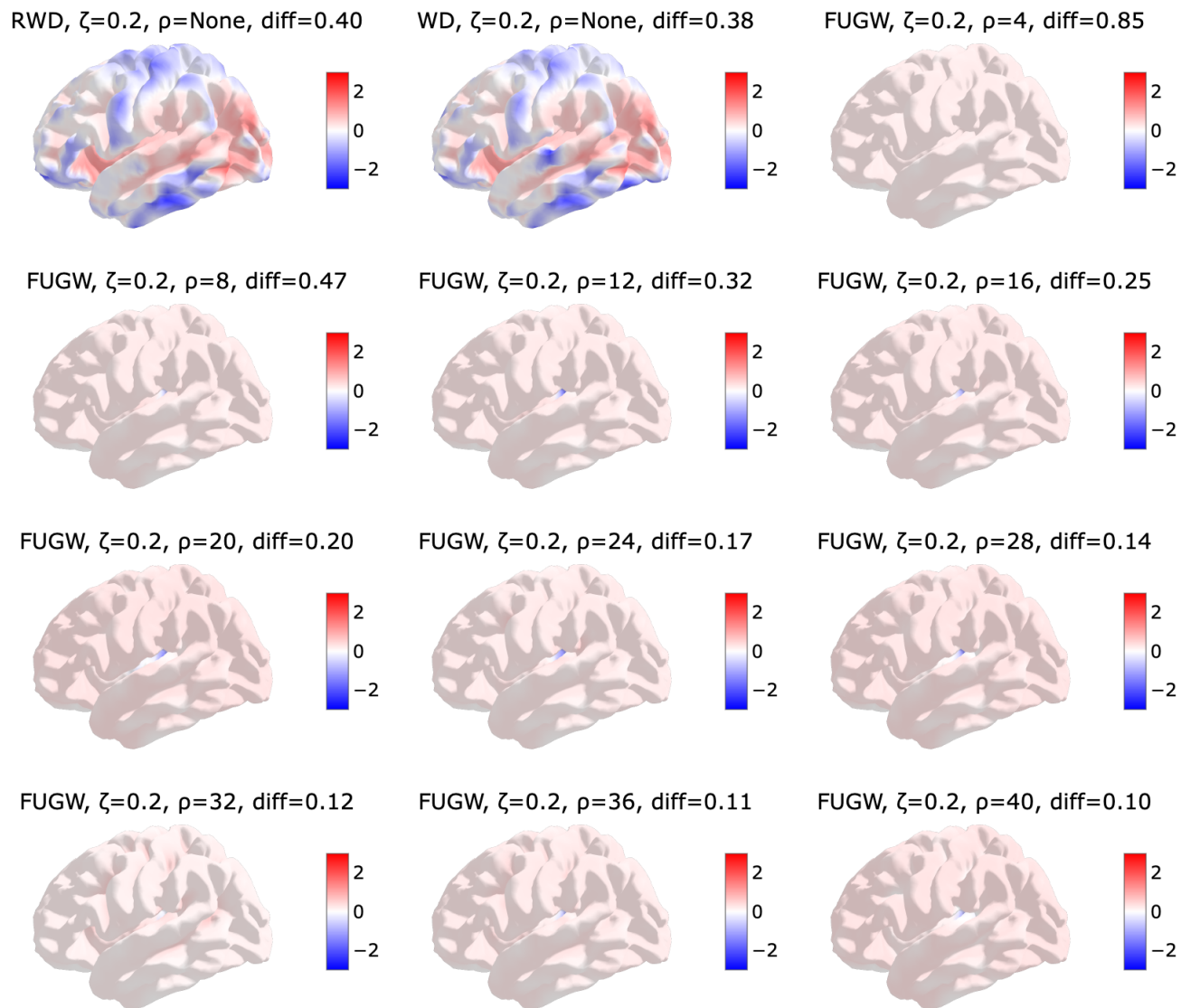


Fig. 6: z-score of barycenters $\zeta = 0.2$ on ADHD dataset.

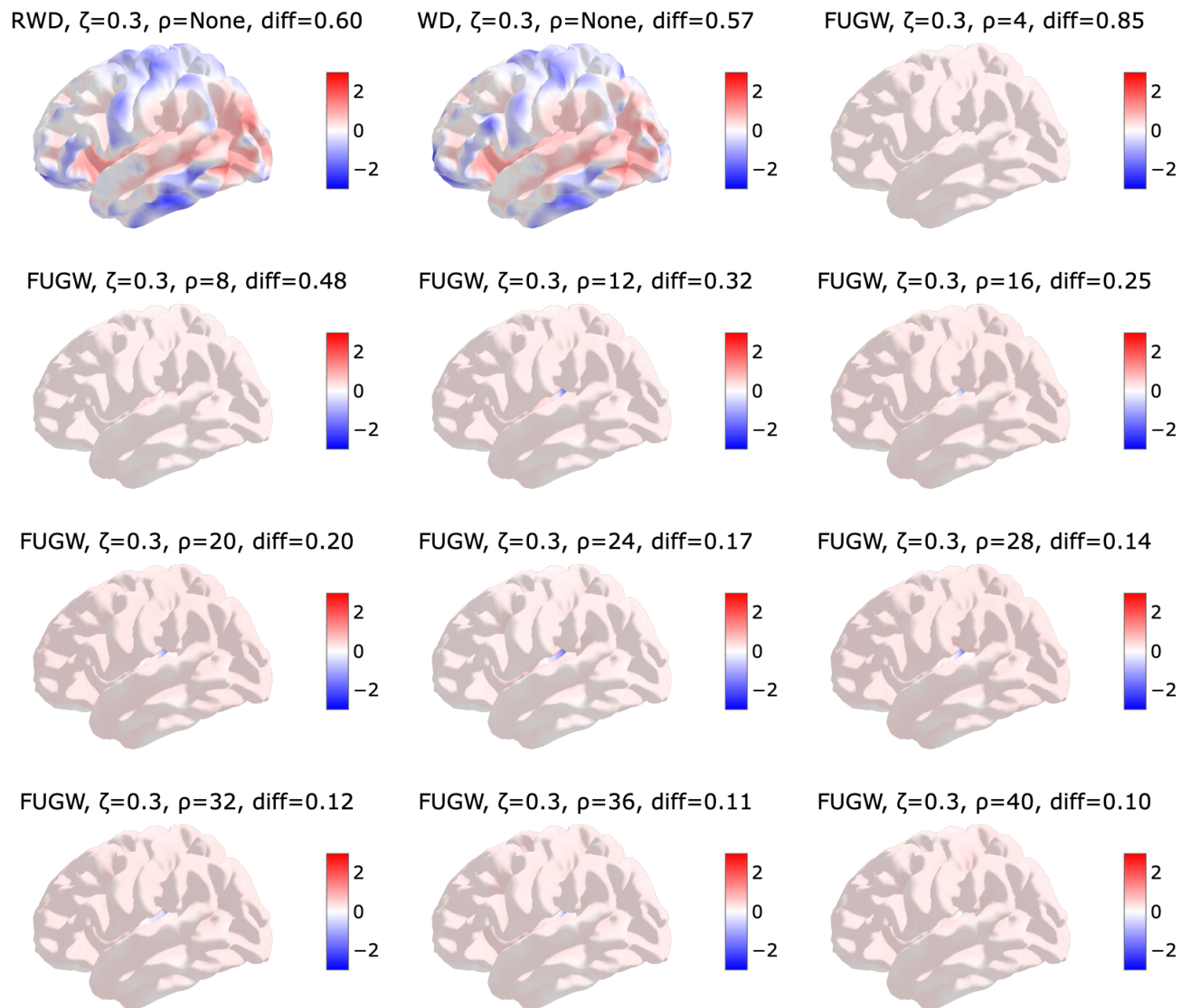


Fig. 7: z-score of barycenters $\zeta = 0.3$ on ADHD dataset.

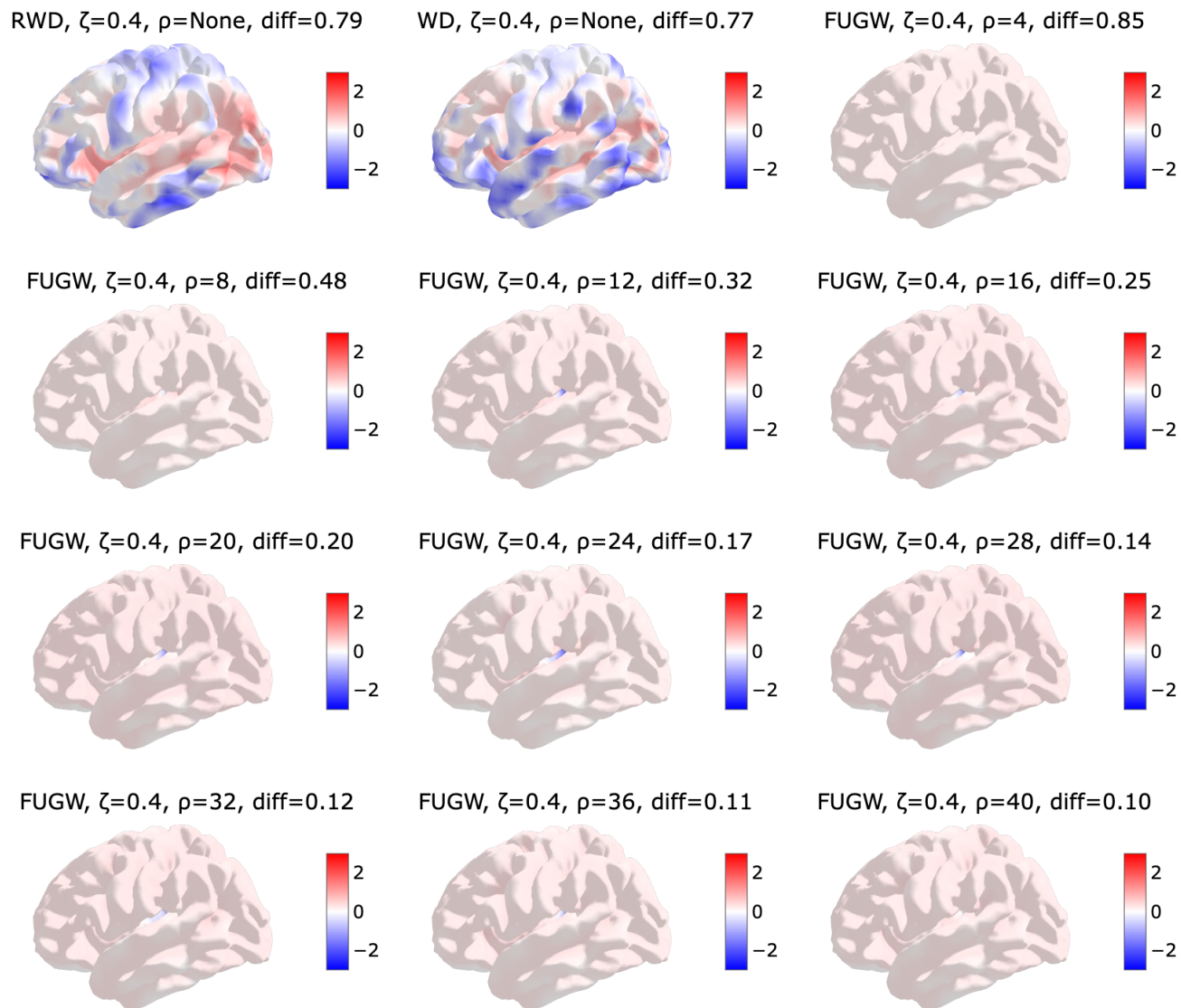


Fig. 8: z-score of barycenters $\zeta = 0.4$ on ADHD dataset.